Ricktho Sarkar

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Education

Indian Institute of Technology (BHU) Varanasi **Nov 2022 - October 2026**
Bachelor of Technology in Electrical Engineering *GPA: 8.19*

Darjeeling Public School **April 2020 - June 2022**
CBSE(XII) *88.40%*

Don Bosco School, Siliguri **April 2020**
ICSE(X) *96.00%*

Technical Skills

Areas of Interest: Robotics, Deep Learning, Computer Vision, Diffusion Models, Natural Language Processing
Languages: C, C++, Python, LaTeX, Java, Dart, GD script, HTML, CSS, Typescript, ReactJS
Tools/Frameworks: Git, Unix Shell, ROS, Carla, PyTorch, Gazebo, OpenCV, Flutter, Detectron2, Jupyter

Work Experience

AI intern| *Field Genie——IOT++——MultipliedAI* 12th May, 2025 - 14th July, 2025

- Worked in developing an **Agentic OS** building multiple agents and integrating them into **LangGraph** and **LangChain**
- Implemented and Built multiple **MCP servers** for automating different tasks of the company
- Created **RAG tools** for different MCP agents along with a sample **Vector Database** in **MongoDB**
- Worked in the backend and frontend using **ReactJS** and **FastAPI** to integrate the Langgraph agents into website.
- **Exposure:** LangGraph, LangChain, PyTorch, ReactJS, Python, MCP, FastAPI

Remote Research Intern | *Indian Statistical Institute, West Bengal, India* 25th November, 2024 - 25th January, 2025

- Learnt about **Vision Language Models** like **CLIP, BLIP, BLIP2** along with their architecture and implementation.
- Learnt about **Normalizing Flows** and its application and usage in different architectures in the recent times.
- Worked on Infusing **Normalizing Flow** and **Variational Auto-Encoder** in VLMs to improve their perfromance.
- Got significant better results than original papers and **publishing** paper for the same.
- **Exposure:** Transformers, Vision Language Models, Variational Autoencoders, Flows

Onsite Research Intern | *Visual AI Lab, Changwon National University, South Korea* 15th May - 15th July, 2024

- Comparative Study in the field of **2D object detection** using **YOLOv8** and other YOLO versions' architecture.
- Implemented the research paper termed as **BEVFusion** by MIT Research group, learning about **Swin Transformer**.
- Studied the **VoxelNet** architecture to process LIDAR Point Cloud Data and convert into BEV feature maps.
- Used the **Detectron2** library by Facebook AI Research Group to implement **Transfer learning** for custom dataset.
- **Exposure:** PyTorch, CNN, Neural Networks, Transformers, Spyder, Jupyter Notebook, CUDA

Projects

High-Resolution Elemental Mapping of the Moon's surface | *Inter-IIT Tech Meet 13.0*

- Developed a method to analyze XRF data from **Chandrayaan-2's CLASS** for elemental mapping of the Moon.
- Analyzed astronomical Time Series Data in **FITS format** with software like **XSPEC** in **Unix-based kernel**
- Applied image processing techniques like **DCT** and **Drizzle** on **structured data** to increase the spatial resolution
- **Exposure:** Python, GoLang, XSPEC, Time Series Data, Image Processing

Autonomous Vehicle | *Team Averera, IIT(BHU)* Sample Video

- Building **SAE Level 3 Autonomous Vehicle** to autonomously navigate using **3D LIDAR and Cameras**.
- Working towards applying **HDL graph SLAM** implementing autonomous navigation using **Pure Pursuit** package.
- Applied **CNN-based** segmentation and trajectory planning, validated in hardware, along with real-time inference
- **Exposure**: Python, Carla, ROS, Robotics Control, PID, 3D Object Detection, Jetson Xavier

Client Selection in Federated Learning | *DRDO Research Project, IIT(BHU)*

- Developed a **client selection** framework leveraging **reinforcement learning** for efficient simulation.
- Optimized outcomes by tuning **privacy budget** and introducing **novel parameters** related to drone clients.
- Enhanced performance by integrating **novel methods** with approaches from complementary research.
- **Exposure:** differential privacy, federated learning, and reinforcement learning.

Agricultural theme manual and autonomous Bot | *Team Robocon 2024, IIT(BHU)*

- Developed Manual and Autonomous Bots for picking seedlings, paddy rice, and empty grain in designated areas.
- Simulated and implemented in **Gazebo ROS** using **2D LIDAR**, **IMU**, cameras and motors with **PID control**.
- Utilized **Jetson Nano** and **Raspberry Pi** and implemented built packages like **Tele-operations** and **SLAM**.
- **Exposure**: Python, Gazebo, ROS, Robotics Control, Segmentation, PID

Memberships

- Joint General Secretary of **Science and Technology Council**, IIT(BHU),led 1000+ multi-domain students.
- Deputy Contingent Leader, **Inter-IIT Tech Meet 14.0**; led 100+ members, IIT (BHU) ranked **4th overall**.
- Technical Lead, **RoboReg**, promoting a research culture in the college and leading research projects.
- Team Lead, **Robocon 2024**, representing IIT(BHU) and coordinating different verticals of the team.
- Technical Coordinator, **Team Averera**, IIT(BHU) Varanasi, building Level-5 autonomous vehicle.
- Under Secretary General, **IIT(BHU) MUN**, 12th edition, organized a 400+ participant conference.
- Student Liaison Officer, **Y20-G20 summit, Varanasi**, aided in hospitality of foreign delegates.

Relevant Coursework

- **MA-102** : Linear Algebra and Derivatives
- **CSO-432**: AI for Programmers
- **CSO-101** : Computer Programming
- **MA-202** : Probability and Statistics
- **CSE-530** : Information Security
- **MA-201** : Numerical Techniques
- **LLG-309**:Academic Writing for Science and Technology
- **8-week Machine Learning Course** by AI Club, IIT Guwahati
- **Summer Analytics 2023 - Data Science and Analytics Course** by Consulting and Analytics Club, IIT Guwahati
- **Completion of Quantum Computing with Qsim** by IIT Roorkee and C-DAC Hyderabad
- **Time Series Analysis Course** by Consulting and Analytics Club, IIT Guwahati
- **Introductory Options with Hands on Programming** by Finance and Economics Club, IIT Guwahati

Achievements

- **National Finalist** in Robocon 2025, after qualifying Stage 1 and 2, out of 100+ colleges.
- **3rd in Cosmic Classifier**, an ML event at **Cognizance'25, IIT Roorkee**, out of 200+ participations.
- **Ranked 4th in all 23 IITs** in **Inter-IIT Tech Meet 13.0** for a High-Prep Problem Statement of ISRO.
- Men-tee at **Amazon ML Summer School 2024**, coming in as the top few in their initial open evaluation.
- Qualified stage 1 OF Robocon 2024, represented IIT(BHU) in Stage 2 and exhibited our bot in Stage 3.
- Honorable mention at **Flipkart Grid 5.0** and was in the top 30 of 100+ participating teams.
- Accomplished Gold Level in **World Quant Brain**, by submitting a certain number of alphas.
- In the top 25 percentile of the one month long Summer Analytics 2023 course on Data Science and Analysis.

Extracurriculars

- Completed 1st year in Vocal Classical and Rabindra Sangeet
- Completed 1st Year in Fine Arts
- Participated in Aagman 2022, college fresher's competition in Fine Art Series and Group Dance